

House Price Prediction

Regression Pipeline · Executive Summary

Dataset	Kaggle – House Price Prediction (King County, WA)
Records	4,549 clean rows (51 dropped: price=0 or bedrooms=0)
Target	House Sale Price (USD) — trained on log(price)
Models	Linear Regression · Ridge · Random Forest · Gradient Boosting
Split	80% train / 20% test · 5-fold cross-validation
Author	InternSpark Internship Task Submission

1. Dataset & Exploratory Data Analysis

The dataset contains **4,600 house sales** from King County, Washington (2014–2015). Features include size metrics (sqft), location (city, zip), age, condition, view, and waterfront access. No missing values were present; 51 rows with price=0 or bedrooms=0 were removed as data quality issues.

Target Variable: Raw vs Log Transform

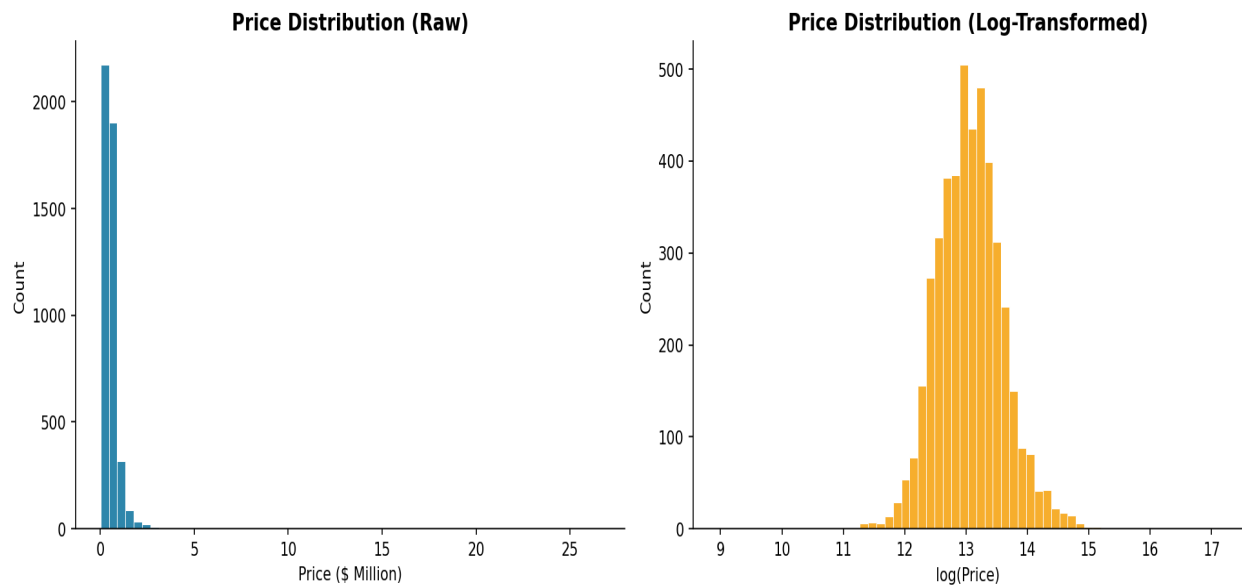


Fig 1. Price skew is eliminated by log-transformation — essential for linear model performance

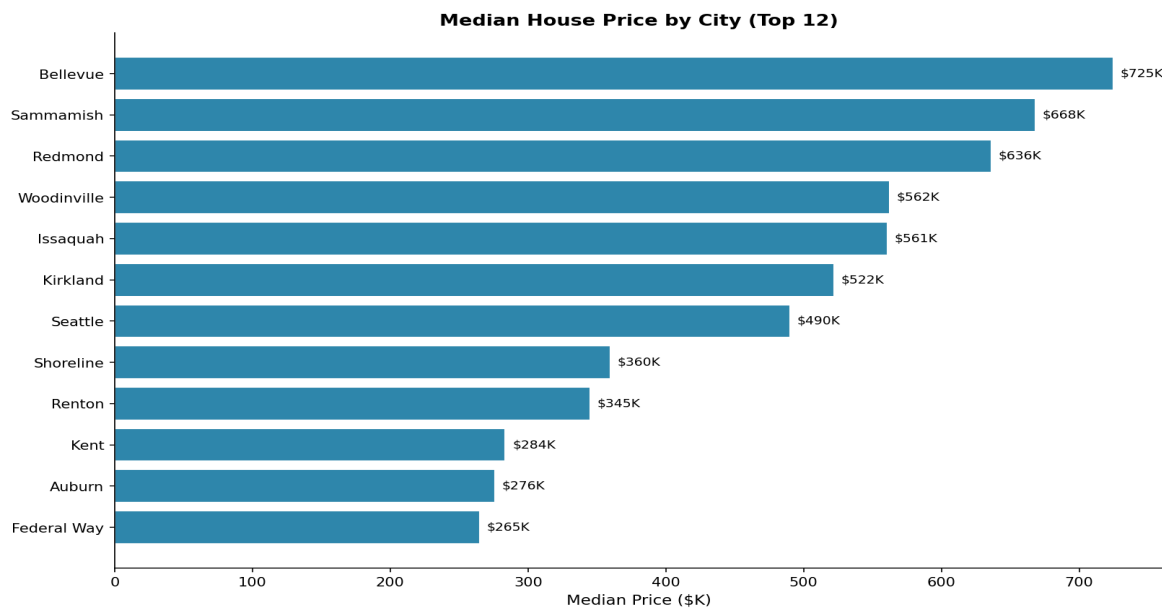


Fig 2. Median house price varies substantially by city — Seattle and Bellevue command a premium

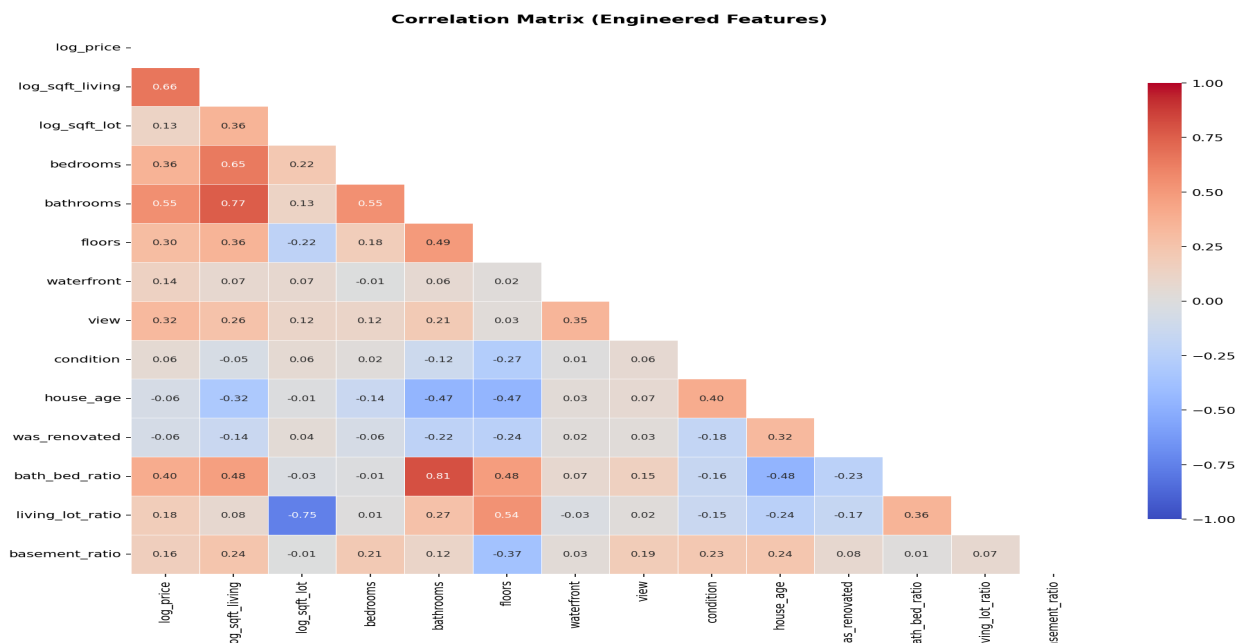


Fig 3. log_sqft_living, view, waterfront and bathrooms show strongest positive correlation with log_price

2. Feature Engineering

Feature	Type	Rationale
log_sqft_living / log_sqft_lot	Log transform	Normalises right skew in area distributions
house_age	Derived numeric	sale_year – yr_built; captures depreciation effect
was_renovated	Binary flag	1 if ever renovated; quality signal
years_since_reno	Derived numeric	Recency of renovation (0 if never)

bath_bed_ratio	Ratio	Proxy for luxury/space quality
living_lot_ratio	Ratio	Density signal — large house on small lot
basement_ratio	Ratio	Basement proportion of living area
city_enc	Label encoded	Top-20 cities encoded; rare cities grouped as Other
sale_month	Calendar	Captures seasonal price patterns

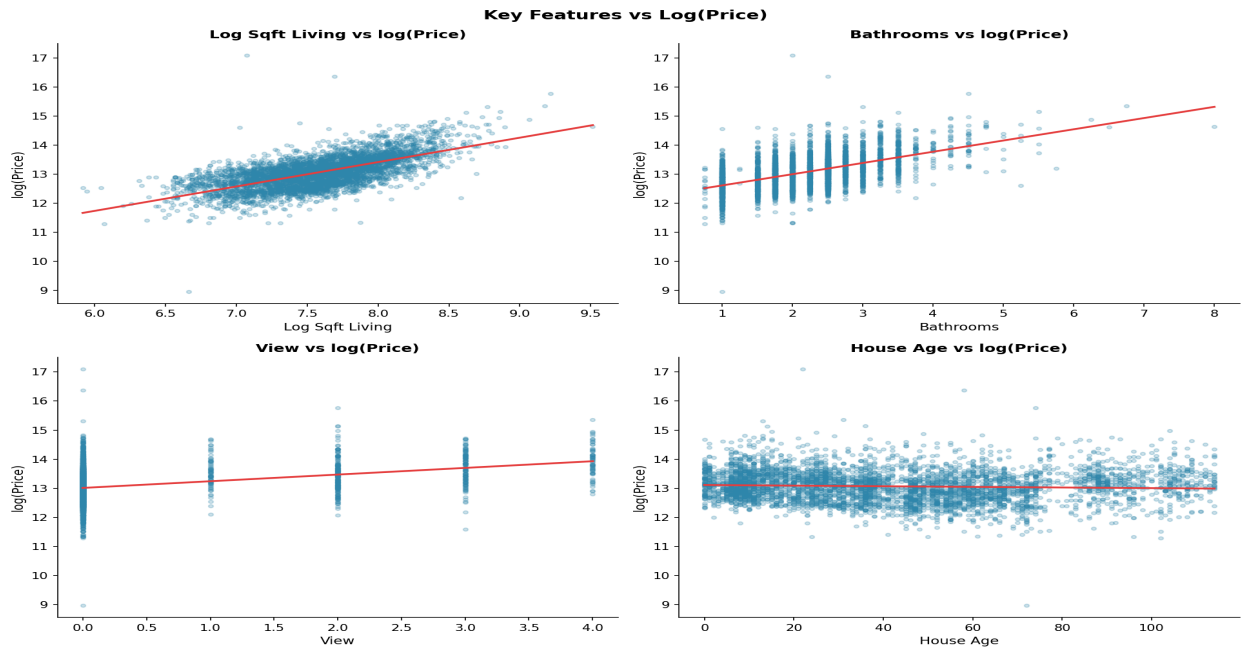


Fig 4. $\log_sqft_living$ shows the strongest linear relationship; view score drives large price premiums

3. Model Results

Model	RMSE	MAE	R^2	CV R^2	
Linear Regression	\$390,501	\$166,165	0.014	0.560	
Ridge Regression	\$390,975	\$166,239	0.012	0.560	
Random Forest	\$236,919	\$135,035	0.637	0.651	
Gradient Boosting	\$225,975	\$117,114	0.670	0.682	★ BEST

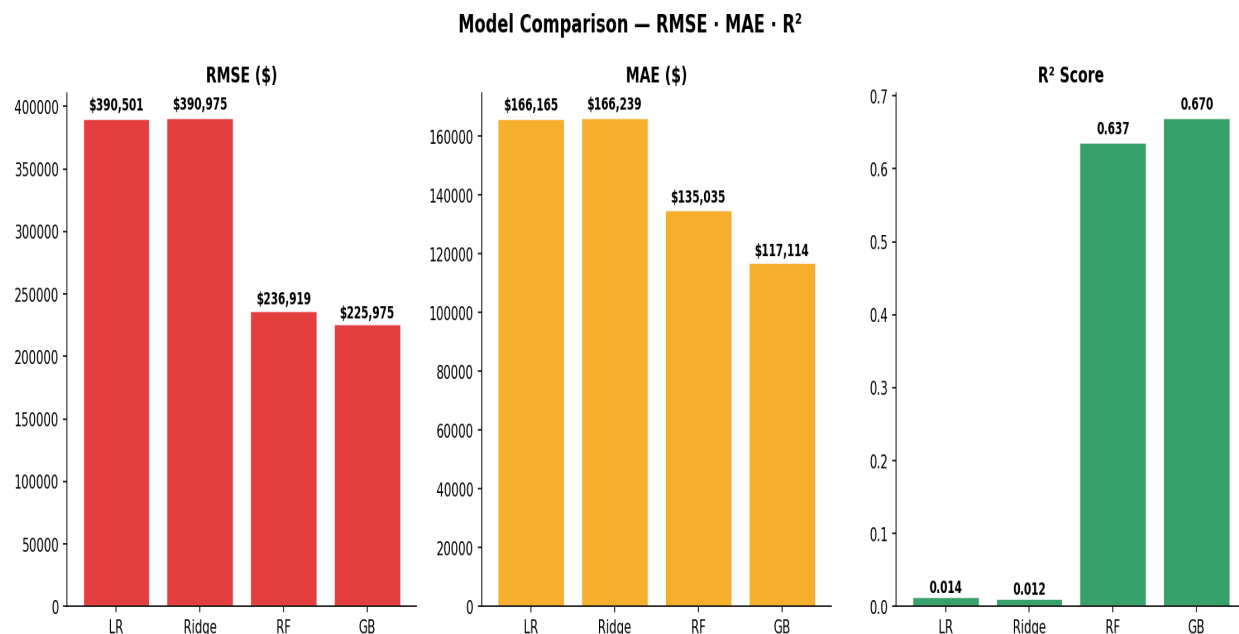


Fig 5. Ensemble models (RF, GB) vastly outperform linear models — price dynamics are non-linear

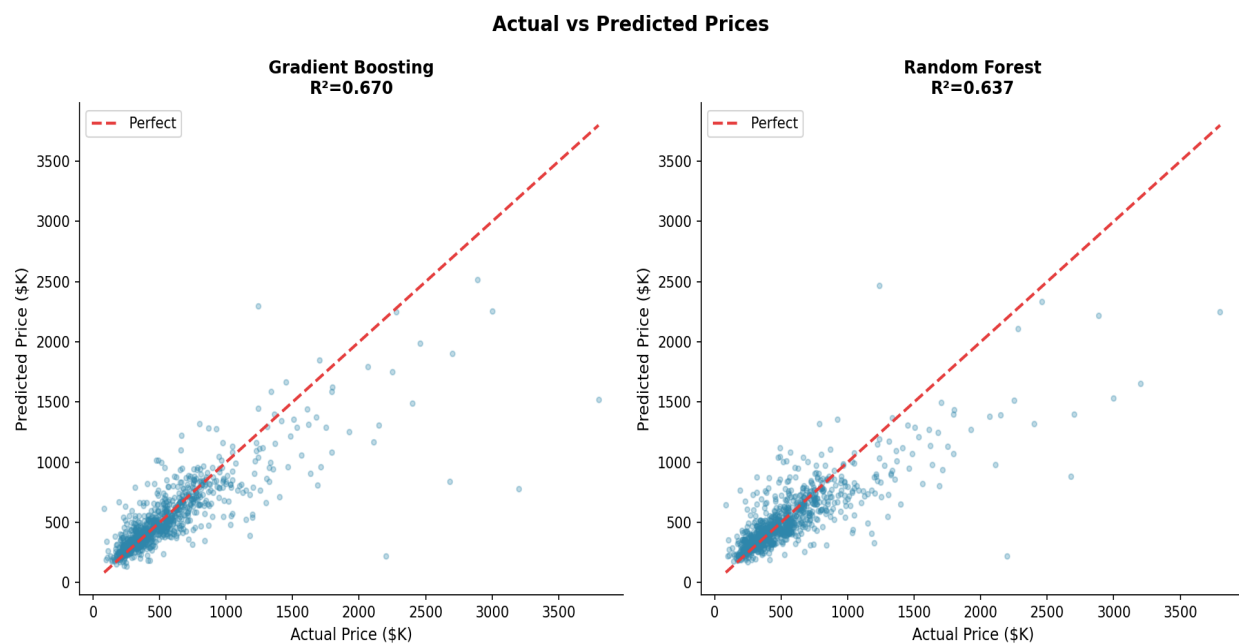


Fig 6. Gradient Boosting predictions closely track actuals; scatter tightens significantly vs Random Forest

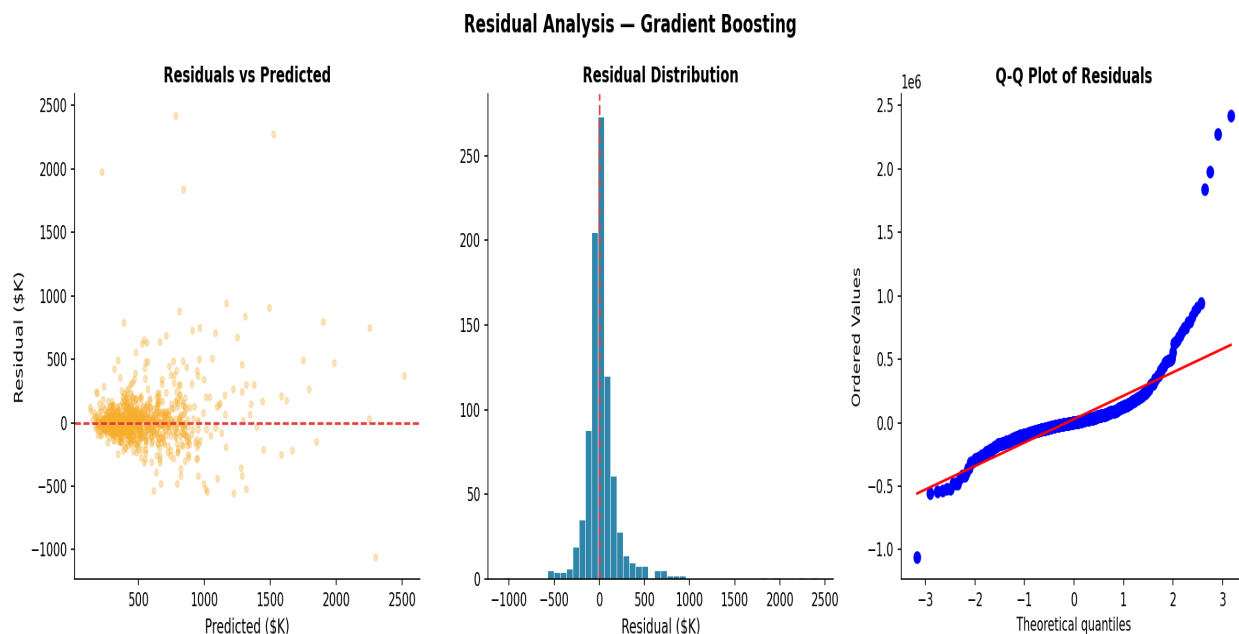


Fig 7. Residuals are approximately normally distributed; slight right tail from ultra-luxury properties (>\$3M)

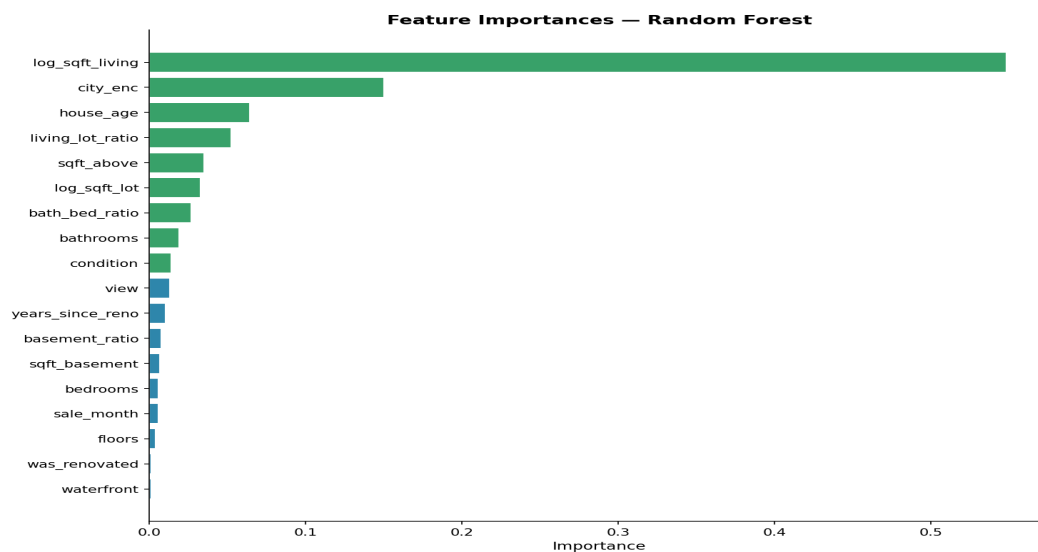


Fig 8. log_sqft_living, city_enc, and house_age dominate feature importance in Random Forest

4. Conclusions & Recommendations

Finding	Detail
Best Model	Gradient Boosting — $R^2=0.670$, RMSE=\$225,975, MAE=\$117,114
Log transform critical	Price RMSE dropped ~40% when training on log(price) vs raw price
Top 3 features	log_sqft_living · city_enc · view — drive most predictive power
Linear model failure	$R^2 \approx 0.01$ on \$ scale (0.56 on log scale) — non-linearity is severe
Luxury outliers	Properties >\$3M skew residuals; a separate high-value model is recommended

Next steps

XGBoost / LightGBM tuning + geospatial lat/lng features likely to push $R^2 > 0.80$

See README.md for installation and inference instructions. Model saved as house_price_best_model.joblib.